Rishabh Shah

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Education

Rutgers University – New Brunswick — Expected May 2027
M.S. in Computer Science, GPA: 3.83/4.0 — *Graduate Student*

University of Mumbai — Dec 2021 – Jun 2025
B.Tech. in Artificial Intelligence & Data Science, CGPA: 7.76/10.0 (Indian scale) — *Mumbai, India*

Technical Skills

Languages: Python, C++, Java, TypeScript, SQL, C
Frameworks & Tools: Django, Next.js, REST APIs, JWT Auth, Docker, Git, GitHub, GitHub Actions (CI/CD), Linux
Databases & Cloud: PostgreSQL, MySQL, MongoDB, AWS, Azure, Google Cloud Platform (GCP)
Testing: Pytest, Vitest, Playwright, Postman
AI/ML: TensorFlow, PyTorch, Scikit-learn, OpenCV, YOLO

Professional Experience

Aaditya Technologies — Onsite – Mumbai, India
Software Engineering Intern — *Jan 2025 – Jul 2025*

- Reduced API response times exceeding 400ms by restructuring MySQL schemas and rewriting critical SQL paths in Python and Next.js, cutting average latency by 22% in Postman API tests
- Eliminated unauthorized cross-service requests across 4 backend services by building JWT-authenticated REST APIs with role-based access control, enforcing identity verification on every internal call
- Containerized 4 backend services with Docker and built AWS release workflows, reducing manual per-release deployment setup by 65% for a 4-person backend team

Creative Line — Onsite – Mumbai, India
Web Development Intern — *Jun 2024 – Sep 2024*

- Resolved a coupling bottleneck across 3 features by refactoring 7 shared-logic components into isolated responsibilities, cutting per-file size by 70% and reducing review complexity
- Traced P95 latency above 3100ms to missing indexes and absent connection pooling, then rebuilt 3 Next.js routes with indexed queries and pooled connections, reducing P95 latency to 420ms
- Closed SQL injection exposure in user-facing forms by switching to parameterized queries and adding input validation across authentication and CRUD endpoints, eliminating direct raw-input-to-database paths

Technical Projects

SnapInterview | AI Interview Practice Platform | snapinterview.app — github.com/RishiiShah/SnapInterview
Next.js, TypeScript, PostgreSQL, Prisma, Groq API, AWS S3, ElevenLabs, Better Auth

- Deployed snapinterview.app as a full-stack interview platform with an async pipeline coordinating microphone capture, LLM streaming, TTS synthesis, and S3 uploads, eliminating audio lag and race conditions
- Resolved a production session-validation issue where route transitions triggered repeated database checks and server crashes, redesigning revocation checks with a 5-minute cache to reduce database load
- Hardened auth and upload flows with per-user S3 key scoping, CSRF protection, rate limits across 4 endpoint types, and CI gates for lint, typecheck, unit, and e2e tests

CseGraph | Repository Context Engine for Coding Agents — github.com/RishiiShah/Csegraph
Python, SQLite, Tree-sitter, FTS5, MCP, VS Code Extension

- Built a local-first repository context engine that parses code with Tree-sitter, stores file/symbol dependency graphs in SQLite, and returns task-specific context for coding agents
- Implemented SHA256 incremental refresh, FTS5 BM25 retrieval, exact-name/path boosting, hub-aware BFS traversal, and deterministic token-budgeted context selection
- Reduced agent context from a 394,638-token repository baseline to 6,564–12,891 retrieved tokens across benchmark runs, cutting token load by 96.73–98.34% while preserving 100% expected-hit coverage across 5 self-corpus tasks

Traffic Violation Detection & Automated Ticketing System — github.com/RishiiShah/Traffic
IEEE ICICIS 2025 | Scopus Index

- Engineered the violation-to-notification pipeline with frame-by-frame detection, trigger-point violation logic, S3 evidence storage, and Twilio SMS alerts, completing processing within source-video duration plus 10 seconds
- Implemented a model benchmarking harness for YOLOv11x/v12x nano-to-XL variants, selecting YOLOv12x at 93.76% end-to-end accuracy with 97.56% vehicle and 93.26% helmet classification
- Replaced traditional OCR that failed on angled low-light plates by integrating Groq Llama-4 Scout as a vision-based fallback with regex post-validation, reaching 92.7% plate extraction accuracy